



Docker & Entornos Cloud.

***Carrera: Ingeniería en Desarrollo de
Software.***

Materia: 27394-CÓMPUTO EN LA NUBE.

Docente: Miguel Angel Cardona Contreras.

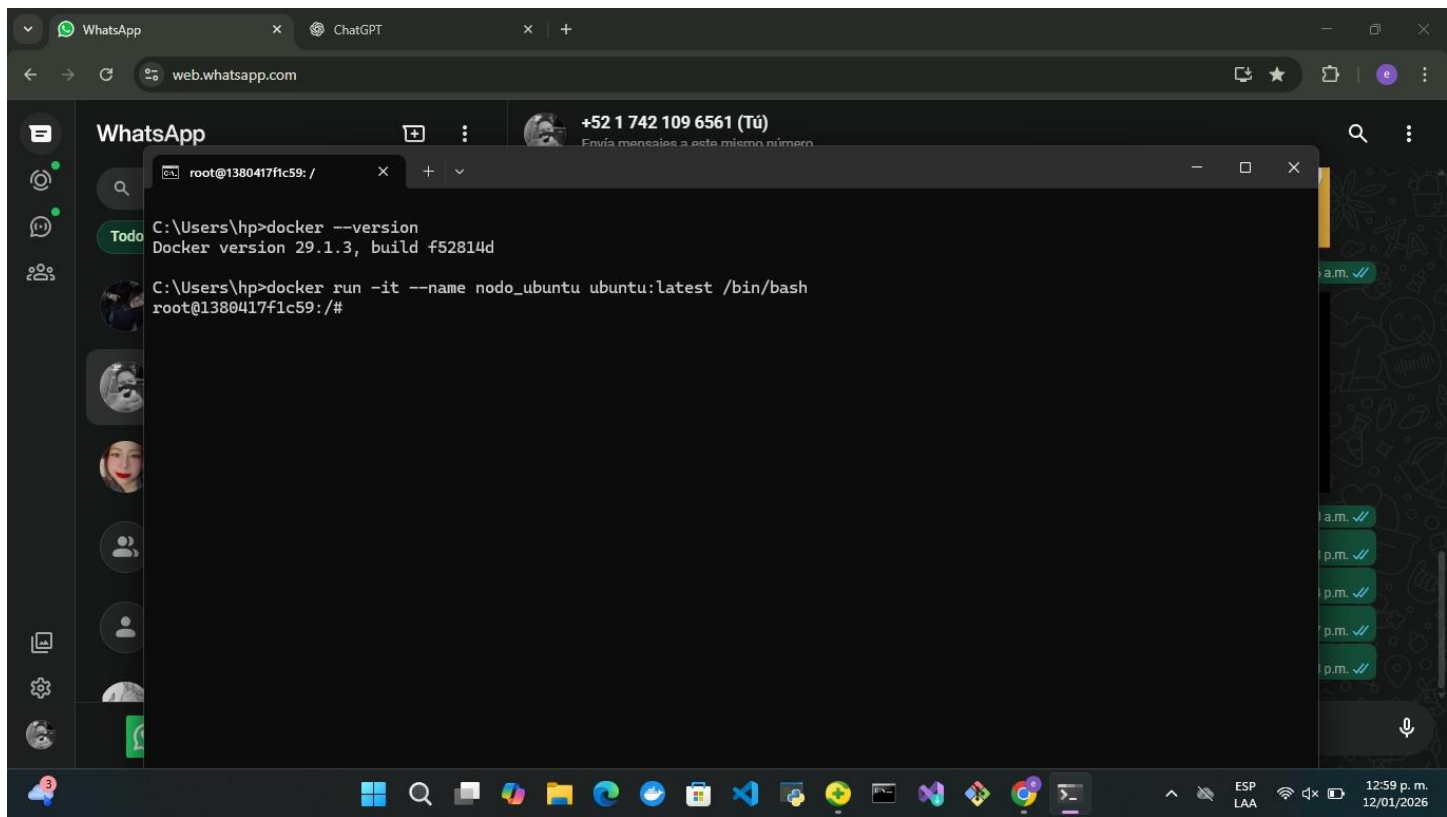
***Modalidad y Turno: Ejecutiva, turno
Vespertino.***

Pino Hernández Edwin Daniel. 2502230111

Buenos días. En esta práctica se utilizó Docker para crear un entorno aislado con Ubuntu, con el objetivo de comprender cómo funcionan los contenedores dentro del cómputo en la nube y preparar el entorno para el proyecto Infoga.

Durante la instalación de Docker Desktop surgió un pequeño obstáculo que me impidió seguir la práctica al pie de la letra durante la clase. Sin embargo, gracias a que las instrucciones estaban bien redactadas en la plataforma, fue posible continuar con la actividad. El problema principal fue la configuración de WSL, que inicialmente presentó errores. Después de actualizar Windows, habilitar correctamente WSL 2 y reiniciar el sistema en varias ocasiones, Docker funcionó correctamente. Una vez resuelto este inconveniente, se desplegó el contenedor de Ubuntu, se instalaron las herramientas necesarias y se clonó el repositorio del proyecto Infoga desde GitHub sin mayores complicaciones.

Docker demostró ser una herramienta eficiente, rápida y menos pesada que una máquina virtual tradicional. A pesar del pequeño inconveniente inicial con WSL, el uso de contenedores facilitó la creación de un entorno limpio y controlado, confirmando que Docker simplifica el despliegue de proyectos y hace el trabajo en la nube mucho más práctico.



```
root@1380417f1c59: /
C:\Users\hp>docker --version
Docker version 29.1.3, build f52814d

C:\Users\hp>docker run -it --name nodo_ubuntu ubuntu:latest /bin/bash
root@1380417f1c59:/# apt update
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble InRelease [256 kB]
Get:3 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1765 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:5 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [33.1 kB]
Get:6 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [2908 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1188 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [331 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble/universe amd64 Packages [19.3 MB]
Get:11 http://archive.ubuntu.com/ubuntu noble/main amd64 Packages [1808 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble/restricted amd64 Packages [117 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages [35.9 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1955 kB]
Get:15 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [3077 kB]
Get:16 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [2141 kB]
Get:17 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Packages [49.5 kB]
Get:18 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [34.6 kB]
Fetched 35.4 MB in 7s (5167 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
11 packages can be upgraded. Run 'apt list --upgradable' to see them.
root@1380417f1c59:/# |
```

```
root@1380417f1c59: /
Please select the geographic area in which you live. Subsequent configuration questions will narrow this down by presenting a list of cities,
representing the time zones in which they are located.

  1. Africa    3. Antarctica  5. Asia    7. Australia  9. Indian   11. Etc
  2. America  4. Arctic    6. Atlantic 8. Europe   10. Pacific 12. Legacy
Geographic area: 2

Please select the city or region corresponding to your time zone.

  1. Adak                33. Cancun            65. Guyana            97. Metlakatla        129. Resolute
  2. Anchorage           34. Caracas           66. Halifax           98. Mexico_City        130. Rio_Branco
  3. Anguilla            35. Cayenne           67. Havana            99. Miquelon           131. Santa_Isabel
  4. Antigua             36. Cayman            68. Hermosillo        100. Moncton           132. Santarem
  5. Araguaia            37. Chicago           69. Indiana/Indianapolis 101. Monterrey         133. Santiago
  6. Argentina/Buenos_Aires 38. Chihuahua         70. Indiana/Knox       102. Montevideo        134. Santo_Domingo
  7. Argentina/Catamarca 39. Ciudad_Juarez     71. Indiana/Marengo    103. Montreal          135. Sao_Paulo
  8. Argentina/Cordoba   40. Coral_Harbour     72. Indiana/Petersburg 104. Montserrat        136. Scoresbysund
  9. Argentina/Jujuy     41. Costa_Rica        73. Indiana/Tell_City  105. Nassau            137. Shiprock
  10. Argentina/La_Rioja  42. Coyhaique         74. Indiana/Vevay      106. New_York           138. Sitka
  11. Argentina/Mendoza  43. Creston           75. Indiana/Vincennes  107. Nipigon           139. St_Barthelemy
  12. Argentina/Rio_Gallegos 44. Cuiaba            76. Indiana/Winamac    108. Nome              140. St_Johns
  13. Argentina/Salta    45. Curacao           77. Inuvik            109. Noronha           141. St_Kitts
  14. Argentina/San_Juan  46. Danmarkshavn      78. Iqaluit           110. North_Dakota/Beulah 142. St_Lucia
  15. Argentina/San_Luis  47. Dawson            79. Jamaica           111. North_Dakota/Center 143. St_Thomas
  16. Argentina/Tucuman   48. Dawson_Creek      80. Juneau            112. North_Dakota/New_Salem 144. St_Vincent
  17. Argentina/Ushuaia   49. Denver            81. Kentucky/Louisville 113. Nuuk              145. Swift_Current
  18. Aruba               50. Detroit           82. Kentucky/Monticello 114. Ojinaga           146. Tegucigalpa
  19. Asuncion            51. Dominica          83. Kralendijk         115. Panama            147. Thule
  20. Atikokan           52. Edmonton          84. La_Paz            116. Pangnirtung        148. Thunder_Bay
  21. Atka               53. Eirunepe          85. Lima              117. Paramaribo         149. Tijuana
  22. Bahia              54. El_Salvador       86. Los_Angeles        118. Phoenix           150. Toronto
  23. Bahia_Banderas     55. Ensenada          87. Lower_Princes     119. Port-au-Prince     151. Tortola
  24. Barbados           56. Fort_Nelson       88. Maceio             120. Port_of_Spain     152. Vancouver
  25. Belem              57. Fortaleza         89. Managua           121. Porto_Acre        153. Virgin
```

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root@1380417f1c59: /
27. Blanc-Sablon        59. Goose_Bay        91. Marigot           123. Puerto_Rico       155. Winnipeg
28. Boa_Vista           60. Grand_Turk       92. Martinique        124. Punta_Arenas     156. Yakutat
29. Bogota              61. Grenada          93. Matamoros         125. Rainy_River      157. Yellowknife
[More] 149

30. Boise               62. Guadeloupe       94. Mazatlan          126. Rankin_Inlet
31. Cambridge_Bay      63. Guatemala        95. Menominee         127. Recife
32. Campo_Grande        64. Guayaquil        96. Merida            128. Regina

Time zone: 94

Current default time zone: 'America/Mazatlan'
Local time is now:      Mon Jan 12 12:04:23 MST 2026.
Universal Time is now:  Mon Jan 12 19:04:23 UTC 2026.
Run 'dpkg-reconfigure tzdata' if you wish to change it.

Setting up libx11-data (2:1.8.7-1build1) ...
Setting up librtmp1:amd64 (2.4+201511223.gitfa8646d.1-2build7) ...
Setting up perl-modules-5.38 (5.38.2-3.2ubuntu0.2) ...
Setting up patch (2.7.6-7build3) ...
Setting up libk5crypto3:amd64 (1.20.1-6ubuntu2.6) ...
Setting up libsasl2-2:amd64 (2.1.28+dfsg1-5ubuntu3.1) ...
Setting up git-man (1:2.43.0-1ubuntu7.3) ...
Setting up netbase (6.4) ...
Setting up libkrb5-3:amd64 (1.20.1-6ubuntu2.6) ...
Setting up libperl5.38t64:amd64 (5.38.2-3.2ubuntu0.2) ...
Setting up libfido2-1:amd64 (1.14.0-1build3) ...
Setting up openssl (3.0.13-0ubuntu3.6) ...
Setting up libbsd0:amd64 (0.12.1-1build1.1) ...
Setting up readline-common (8.2-4build1) ...
Setting up publicsuffix (20231001.0357-0.1) ...
Setting up libldap2:amd64 (2.6.7+dfsg-1~exp1ubuntu8.2) ...
Setting up libxdmcp6:amd64 (1:1.1.3-0ubuntu6) ...
Setting up libxcb1:amd64 (1.15-1ubuntu2) ...
Setting up libedit2:amd64 (3.1-20230828-1build1) ...
```



```
root@1380417f1c59: /
Updating certificates in /etc/ssl/certs...
146 added, 0 removed; done.
Setting up perl (5.38.2-3.2ubuntu0.2) ...
Setting up libgssapi-krb5-2:amd64 (1.20.1-6ubuntu2.6) ...
Setting up libssh-4:amd64 (0.10.6-2ubuntu0.2) ...
Setting up libreadline8t64:amd64 (8.2-4build1) ...
Setting up libx11-6:amd64 (2:1.8.7-1build1) ...
Setting up libxmu1:amd64 (2:1.1.3-3build2) ...
Setting up libpython3.12-stdlib:amd64 (3.12.3-1ubuntu0.10) ...
Setting up openssh-client (1:9.6p1-3ubuntu13.14) ...
Setting up python3.12 (3.12.3-1ubuntu0.10) ...
Setting up libcurl3t64-gnutls:amd64 (8.5.0-2ubuntu10.6) ...
Setting up libxext6:amd64 (2:1.3.4-1build2) ...
Setting up liberror-perl (0.17029-2) ...
Setting up git (1:2.43.0-1ubuntu7.3) ...
Setting up xauth (1:1.1.2-1build1) ...
Setting up libpython3-stdlib:amd64 (3.12.3-0ubuntu2.1) ...
Setting up python3 (3.12.3-0ubuntu2.1) ...
running python rtupdate hooks for python3.12...
running python post-rtupdate hooks for python3.12...
Processing triggers for libc-bin (2.39-0ubuntu8.6) ...
Processing triggers for ca-certificates (20240203) ...
Updating certificates in /etc/ssl/certs...
0 added, 0 removed; done.
Running hooks in /etc/ca-certificates/update.d...
done.
root@1380417f1c59: /# git clone https://github.com/MikeCardona076/Infoga
Cloning into 'Infoga'...
remote: Enumerating objects: 138, done.
remote: Counting objects: 100% (32/32), done.
remote: Compressing objects: 100% (10/10), done.
remote: Total 138 (delta 24), reused 22 (delta 22), pack-reused 106 (from 1)
Receiving objects: 100% (138/138), 5.35 MiB | 1.17 MiB/s, done.
Resolving deltas: 100% (52/52), done.
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```

```
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root@1380417f1c59: /# ls
Infoga bin boot dev etc home lib lib64 media mnt opt proc root run sbin srv sys tmp usr var
root@1380417f1c59: /# exit
```

```
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root@1380417f1c59: /# ls
Infoga bin boot dev etc home lib lib64 media mnt opt proc root run sbin srv sys tmp usr var
root@1380417f1c59: /# exit
exit
C:\Users\hp>docker start -i nodo_ubuntu
root@1380417f1c59: /#
```